SYSTEME D’ANALYSE DE LOGS POUR CYBERSECURITY

BIGLOGS

Présenté par :
BOUALI Youssef
CHAKIR Brahim
OULKADI Mohamed

Encadré par :
Pr. AIT LAHCEN Yassine

Plan

Problématique

Les réseaux modernes génèrent d'énormes volumes de trafic.
Les équipes de sécurité rencontrent des difficultés :

- Détection instantanée
- Analyse en temps réel
- Classification des attaques

Objectif du projet

Construire un système scalable capable de :

- Capturer et analyser le trafic en temps réel
- Classifier les paquets : bénin / attaque
- Stocker les résultats dans une base distribuée
- Offrir une visualisation en temps réel des menaces

Flux de Données Global

Capture Réseau

Scapy capture les paquets et les convertit en format NetFlow enrichi (CSV).

Traitement par Spark

Nettoyage, feature engineering et prédictions ML en temps réel sur les données de flux.

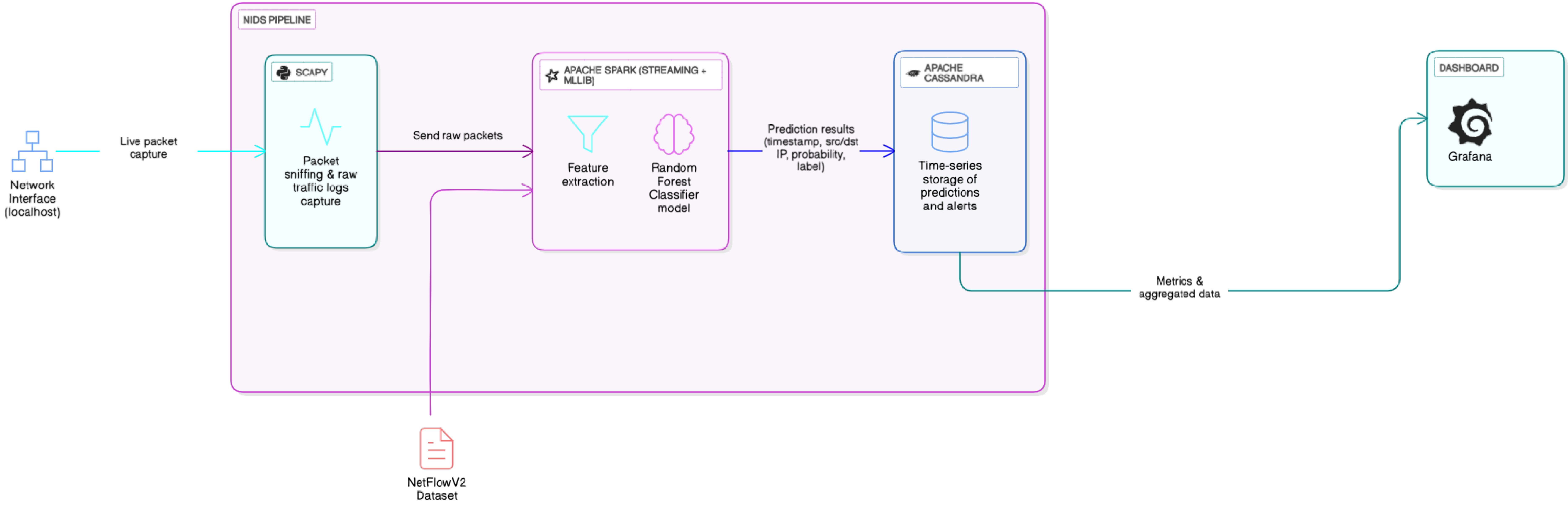
Stockage & ML

Stockage persistant dans Cassandra et application du modèle Random Forest pour la détection.

Visualisation & Alertes

Export des résultats pour dashboards (Grafana) et génération d'alertes.

Architecture de system



Technologies

kaggle

scapy

APACHE Spark™

Grafana

Azure

cassandra

Stack Technologique

Composant	Technologie	Rôle
Orchestration	Docker Compose	5 services conteneurisés
Traitement	Apache Spark 3.x	Jobs distribués PySpark
Stockage	Apache Cassandra 4.1	NoSQL haute performance
ML	Random Forest	80 arbres, profondeur 12
Capture	Scapy	Analyse paquets temps réel
Dataset	NetFlow v9	19M+ flux (Kaggle)

Dataset – NetFlow V2

The University of Queensland
Australia

Home / Marius / NIDS Datasets

Machine Learning-Based NIDS Datasets

The datasets on this page are designed for machine learning-based Network Intrusion Detection Systems (NIDS) and are organised into the following high-level collections:

- **NetFlow V3 Datasets**: This collection consists of four datasets in NetFlow format. This is the only version incorporating **temporal features**. These datasets build on the 43 features of V2 by adding **10 temporal NetFlow features**, totaling 53 features. They are provided in CSV format.
- **NetFlow V2 Datasets**: Comprising five datasets in NetFlow format, this collection extends the V1 datasets with 43 enhanced NetFlow features. They are available in CSV format.
- **NetFlow V1 Datasets**: This collection includes five datasets converted from four diverse formats into NetFlow format, featuring 12 basic NetFlow features. They are provided in CSV format.
- **CICFlowMeter Datasets**: Consisting of five datasets, this collection is generated using the CICFlowMeter tool by converting three datasets from various formats into CICFlowMeter format, with 80 features. They are available in CSV format.

For full details and the recommended citation for these datasets, please refer to the corresponding links in the tabs below.

Dataset – NetFlow V2

Caractéristiques principales:

- Format : NetFlow (CSV)
- Nombre de caractéristiques : 43 features NetFlow avancés
- Type de données : Flots réseau (Network Flows)
- Volume : Plus de 2,3 millions de flots

Pipeline de Machine Learning

Raw Data
43 features NetFlow

VectorAssembler
Combine les features

Random Forest
80 arbres, prof. 12

Prédiction
BENIGN / ATTACK

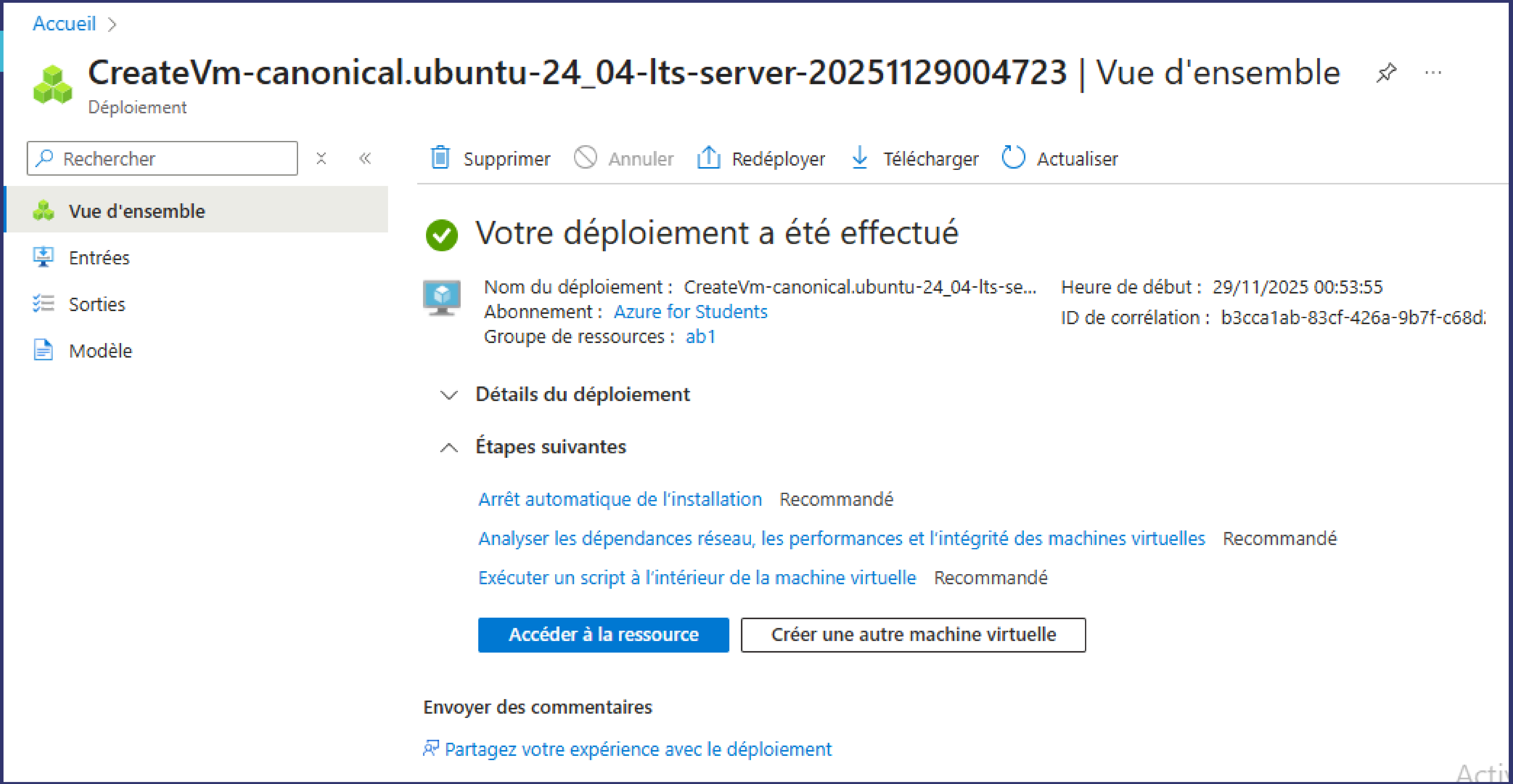
Le modèle **Random Forest** offre est robuste aux datasets déséquilibrés. Il fournit également un **score de confiance** pour chaque prédiction.

MODÈLE ENTRAÎNÉ : CHOIX

Le modèle choisi est un Random Forest Classifier, entraîné avec Spark MLlib, car il offre :

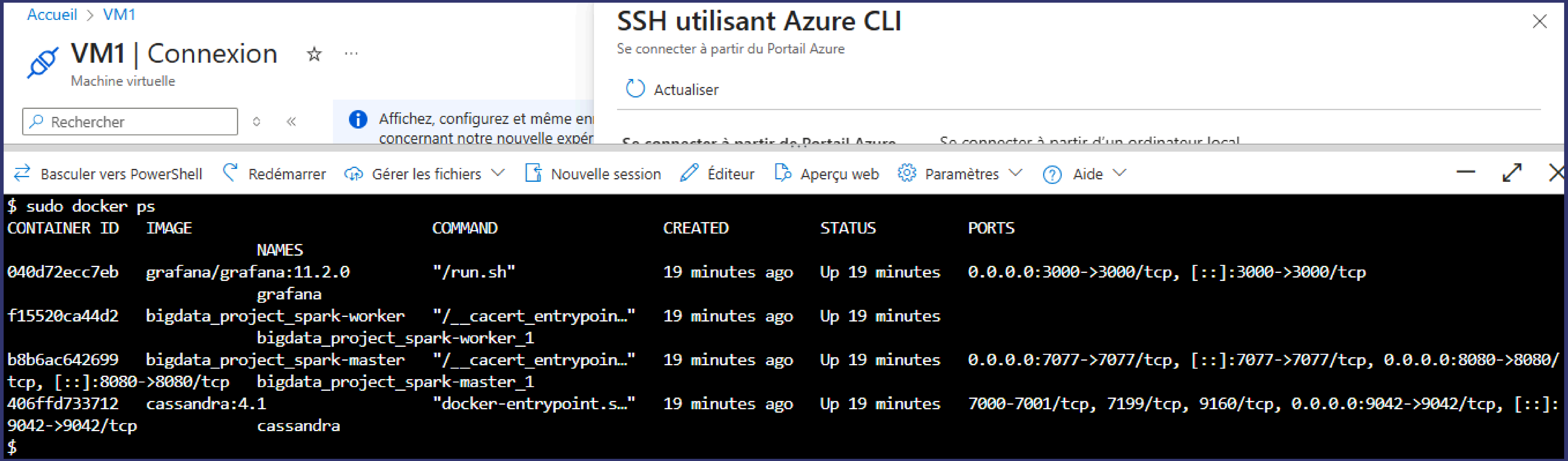
- une excellente précision
- une robustesse face aux données bruitées
- une rapidité d'inférence compatible avec les systèmes temps réel
- une bonne capacité de généralisation sur des flux réseau variés

Déploiement Azure



Infrastructure Dockerisée

Le déploiement est entièrement conteneurisé via docker-compose pour garantir l'isolation, la portabilité et facilite le déploiement sur n'importe quel environnement.



Démonstration & Résultats

Prétraitement et Division Train / Test

```python
# --- 4. Prétraitement des données ---
# Suppression des colonnes non pertinentes pour le modèle
cols_to_drop = ["IPV4_SRC_ADDR", "IPV4_DST_ADDR", "Label"]  # Label redondant avec Attack
df_clean = df.drop(*cols_to_drop)

# Gestion des valeurs manquantes
print("Valeurs manquantes par colonne :")
missing_counts = df_clean.select([_sum(col(c).isNull().cast("int")).alias(c) for c in df_clean.columns])
missing_counts.show()

# Suppression des lignes avec valeurs manquantes
df_clean = df_clean.na.drop()
print(f"Nombre de lignes après nettoyage : {df_clean.count():,}")

# --- 5. Création de la cible binaire (Label_Binaire) ---
df_labeled = df_clean.withColumn(
    "Label_Binaire",
    when(col("Attack") == "Benign", 0).otherwise(1)
)

print("Distribution des classes :")
df_labeled.groupBy("Label_Binaire").count().show()
```

```python
# --- 7. Sélection des features numériques ---
feature_columns = [c for c in df_labeled.columns if c not in ["Attack", "Label_Binaire"]]
print(f"Nombre de features sélectionnées : {len(feature_columns)}")

# Vérification des types
non_numeric = [c for c in feature_columns
               if df_labeled.schema[c].dataType.typeName() not in ["integer", "long", "double", "float"]]
if non_numeric:
    print("Attention : colonnes non numériques détectées :", non_numeric)
else:
    print("Toutes les features sont numériques → OK")

# --- 8. Assemblage des features ---
assembler = VectorAssembler(inputCols=feature_columns, outputCol="features", handleInvalid="skip")
df_assembled = assembler.transform(df_labeled)
df_final = df_assembled.select("features", "Label_Binaire")

print("Exemple de vecteur de features :")
df_final.show(5, truncate=False)

# --- 9. Division Train / Test ---
train_df, test_df = df_final.randomSplit([0.8, 0.2], seed=42)
print(f"Taille entraînement : {train_df.count():,}")
print(f"Taille test : {test_df.count():,}")
```

CLEAN & LOAD

```
youssef@wsoum:~/BigData_Project$ docker exec -it bigdata_project_spark-master_1 spark-submit --master spark://spark-master:7077 --packages com.datastax.spark:spark-cass
andra-connector_2.12:3.4.1 --conf spark.cassandra.connection.host=cassandra --conf spark.cassandra.auth.username=cassandra --conf spark.cassandra.auth.password=cassandr
a /scripts/clean_and_load.py
:: loading settings :: url = jar:file:/opt/spark/jars/ivy-2.5.0.jar!/org/apache/ivy/core/settings/ivysettings.xml
Ivy Default Cache set to: /root/.ivy2/cache
The jars for the packages stored in: /root/.ivy2/jars
com.datastax.spark#spark-cassandra-connector_2.12 added as a dependency
:: resolving dependencies :: org.apache.spark#spark-submit-parent-302c7196-b471-4b1e-a207-4842b2be62ea;1.0
        confs: [default]
        found com.datastax.spark#spark-cassandra-connector_2.12;3.4.1 in central
        found com.datastax.spark#spark-cassandra-connector-driver_2.12;3.4.1 in central
        found org.scala-lang.modules#scala-collection-compat_2.12;2.11.0 in central
        found com.datastax.oss#java-driver-core-shaded;4.13.0 in central
        found com.datastax.oss#native-protocol;1.5.0 in central
        found com.datastax.oss#java-driver-shaded-guava;25.1-jre-graal-sub-1 in central
        found com.typesafe#config;1.4.1 in central
        found org.slf4j#slf4j-api;1.7.26 in central
        found io.dropwizard.metrics#metrics-core;4.1.18 in central
        found org.hdrhistogram#HdrHistogram;2.1.12 in central
        found org.reactivestreams#reactive-streams;1.0.3 in central
        found com.github.stephenc.jcip#jcip-annotations;1.0-1 in central
        found com.github.spotbugs#spotbugs-annotations;3.1.12 in central
        found com.google.code.findbugs#jsr305;3.0.2 in central
        found com.datastax.oss#java-driver-mapper-runtime;4.13.0 in central
        found com.datastax.oss#java-driver-query-builder;4.13.0 in central
        found org.apache.commons#commons-lang3;3.10 in central
        found com.thoughtworks.paranamer#paranamer;2.8 in central
        found org.scala-lang#scala-reflect;2.12.11 in central
downloading https://repo1.maven.org/maven2/com/datastax/spark/spark-cassandra-connector_2.12/3.4.1/spark-cassandra-connector_2.12-3.4.1.jar ...
        [SUCCESSFUL ] com.datastax.spark#spark-cassandra-connector_2.12;3.4.1!spark-cassandra-connector_2.12.jar (396ms)
downloading https://repo1.maven.org/maven2/com/datastax/spark/spark-cassandra-connector-driver_2.12/3.4.1/spark-cassandra-connector-driver_2.12-3.4.1.jar ...
        [SUCCESSFUL ] com.datastax.spark#spark-cassandra-connector-driver_2.12;3.4.1!spark-cassandra-connector-driver_2.12.jar (184ms)
downloading https://repo1.maven.org/maven2/org/scala-lang/modules/scala-collection-compat_2.12/2.11.0/scala-collection-compat_2.12-2.11.0.jar ...
        [SUCCESSFUL ] org.scala-lang.modules#scala-collection-compat_2.12;2.11.0!scala-collection-compat_2.12.jar (134ms)
```

QUERY CASSANDRA : FLOWS

```
youssef@wsoum:~/BigData_Project$ docker exec -it cassandra cqlsh cassandra 9042 -e "SELECT * FROM netflow.flows LIMIT 10;"

 src_ip_int | flow_start_time                 | flow_id | attack_type            | client_tcp_flags | dst_ip        | dst_ip_int | dst_port | dst_to_src_avg_throughput
| duration_ms | in_bytes | in_pkts | l7_proto | label | num_pkts_1024_to_1514_bytes | num_pkts_128_to_256_bytes | num_pkts_256_to_512_bytes | num_pkts_512_to_1024_bytes
 | num_pkts_up_to_128_bytes | out_bytes | out_pkts | protocol | server_tcp_flags | src_ip          | src_port | src_to_dst_avg_throughput | tcp_flags
------------+---------------------------------+---------+------------------------+------------------+---------------+------------+----------+--------------------------
+-------------+----------+---------+----------+-------+-----------------------------+---------------------------+---------------------------+---------------------------
-+--------------------------+-----------+----------+----------+------------------+-----------------+----------+---------------------------+-----------
          0 | 2025-11-19 22:37:58.699000+0000 |       0 |         SSH-Bruteforce |               27 |  172.31.69.25 |          0 |       22 |                3.012e+07
|           0 |     3164 |      23 |       92 |     1 |                           1 |                         7 |                         1 |                          2
 |                       33 |      3765 |       21 |        6 |               27 |     13.58.98.64 |    40894 |                2.5312e+07 |        27
          0 | 2025-11-19 22:37:58.699000+0000 |       1 |                 Benign |              219 | 172.31.66.103 |          0 |     3389 |               1.6248e+07
|           0 |     1919 |      14 |        0 |     1 |                           1 |                         6 |                         0 |                          1
 |                       17 |      2031 |       11 |        6 |               30 | 213.202.230.143 |    29622 |                1.5352e+07 |       223
          0 | 2025-11-19 22:37:58.699000+0000 |       2 |                 Benign |                0 |    172.31.0.2 |          0 |       53 |                1.184e+06
|           0 |      116 |       2 |        0 |     1 |                           0 |                         0 |                         0 |                          0
 |                        4 |       148 |        2 |       17 |                0 |     172.31.66.5 |    65456 |                  9.28e+05 |         0
          0 | 2025-11-19 22:37:58.699000+0000 |       3 |                 Benign |                0 |    172.31.0.2 |          0 |       53 |                 1.04e+06
|           0 |       70 |       1 |        0 |     1 |                           0 |                         1 |                         0 |                          0
 |                        1 |       130 |        1 |       17 |                0 |    172.31.64.92 |    57918 |                   5.6e+05 |         0
          0 | 2025-11-19 22:37:58.699000+0000 |       4 | DDoS attacks-LOIC-HTTP |              222 |  172.31.69.25 |          0 |       80 |                9.088e+06
|     4294827 |      232 |       5 |        7 |     1 |                           0 |                         0 |                         0 |                          1
 |                        8 |      1136 |        4 |        6 |               27 |    18.219.32.43 |    63269 |                      8000 |       223
          0 | 2025-11-19 22:37:58.699000+0000 |       5 |                 Benign |              223 |   23.36.33.52 |          0 |      443 |               2.5352e+07
|           0 |     3988 |      41 |        0 |     1 |                           0 |                         1 |                         3 |                          4
 |                       73 |      3169 |       40 |        6 |               30 |    172.31.65.93 |    51677 |                3.1904e+07 |       223
          0 | 2025-11-19 22:37:58.699000+0000 |       6 |                 Benign |                0 |    172.31.0.2 |          0 |       53 |                 8.48e+05
|           0 |       68 |       1 |        0 |     1 |                           0 |                         0 |                         0 |                          0
 |                        2 |       106 |        1 |       17 |                0 |    172.31.68.18 |    49190 |                  5.44e+05 |         0
          0 | 2025-11-19 22:37:58.699000+0000 |       7 |                 Benign |                0 |    172.31.0.2 |          0 |       53 |                1.864e+06
|           0 |      105 |       1 |    5.178 |     1 |                           0 |                         1 |                         0 |                          0
 |                        1 |       233 |        1 |       17 |                0 |   172.31.65.118 |    51812 |                   8.4e+05 |         0
          0 | 2025-11-19 22:37:58.699000+0000 |       8 |        DDOS attack-HOIC |              219 |  172.31.69.28 |          0 |       80 |                2.288e+06
|     4294963 |      522 |       5 |        7 |     1 |                           0 |                         0 |                         1 |                          1
 |                        8 |      1147 |        5 |        6 |               27 |    18.219.32.43 |    54574 |                  1.04e+06 |       219
```

ENTRAÎNEMENT : SPARK SUBMIT

```
youssef@wsoum:~/BigData_Project$ docker exec -it bigdata_project_spark-master_1 spark-submit --master spark://spark-master:7077 --packages com.datastax.spark:spark-cass
andra-connector_2.12:3.4.1 --driver-memory 4g --executor-memory 2g /scripts/train_model.py
:: loading settings :: url = jar:file:/opt/spark/jars/ivy-2.5.0.jar!/org/apache/ivy/core/settings/ivysettings.xml
Ivy Default Cache set to: /root/.ivy2/cache
The jars for the packages stored in: /root/.ivy2/jars
com.datastax.spark#spark-cassandra-connector_2.12 added as a dependency
:: resolving dependencies :: org.apache.spark#spark-submit-parent-b2485595-d64d-48f5-8611-268bbb718807;1.0
        confs: [default]
        found com.datastax.spark#spark-cassandra-connector_2.12;3.4.1 in central
        found com.datastax.spark#spark-cassandra-connector-driver_2.12;3.4.1 in central
        found org.scala-lang.modules#scala-collection-compat_2.12;2.11.0 in central
        found com.datastax.oss#java-driver-core-shaded;4.13.0 in central
        found com.datastax.oss#native-protocol;1.5.0 in central
        found com.datastax.oss#java-driver-shaded-guava;25.1-jre-graal-sub-1 in central
        found com.typesafe#config;1.4.1 in central
        found org.slf4j#slf4j-api;1.7.26 in central
        found io.dropwizard.metrics#metrics-core;4.1.18 in central
        found org.hdrhistogram#HdrHistogram;2.1.12 in central
        found org.reactivestreams#reactive-streams;1.0.3 in central
        found com.github.stephenc.jcip#jcip-annotations;1.0-1 in central
        found com.github.spotbugs#spotbugs-annotations;3.1.12 in central
        found com.google.code.findbugs#jsr305;3.0.2 in central
        found com.datastax.oss#java-driver-mapper-runtime;4.13.0 in central
        found com.datastax.oss#java-driver-query-builder;4.13.0 in central
        found org.apache.commons#commons-lang3;3.10 in central
        found com.thoughtworks.paranamer#paranamer;2.8 in central
        found org.scala-lang#scala-reflect;2.12.11 in central
:: resolution report :: resolve 3617ms :: artifacts dl 48ms
        :: modules in use:
        com.datastax.oss#java-driver-core-shaded;4.13.0 from central in [default]
        com.datastax.oss#java-driver-mapper-runtime;4.13.0 from central in [default]
        com.datastax.oss#java-driver-query-builder;4.13.0 from central in [default]
        com.datastax.oss#java-driver-shaded-guava;25.1-jre-graal-sub-1 from central in [default]
        com.datastax.oss#native-protocol;1.5.0 from central in [default]
        com.datastax.spark#spark-cassandra-connector-driver_2.12;3.4.1 from central in [default]
```

```
youssef@wsoum:~/BigData_Project$ docker exec -it bigdata_project_spark-master_1 ls -la /models/
total 12
drwxr-xr-x 3 root root 4096 Nov 20 01:13 .
drwxr-xr-x 1 root root 4096 Nov 20 01:13 ..
drwxr-xr-x 4 root root 4096 Nov 20 01:13 2anomaly_detection_model_rf.spark
```

LANCEMENT DE MODÈLE

```
youssef@wsoum:~/BigData_Project$ docker exec -it bigdata_project_spark-master_1 sh -c "cd /scripts && python3 realtime_netflow_predictor.py"
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
25/11/23 10:36:48 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Loading model from /model/anomaly_detection_model_rf.spark ...
WARNING: An illegal reflective access operation has occurred        (0 + 1) / 1]
WARNING: Illegal reflective access by org.apache.spark.util.SizeEstimator$ (file:/opt/spark/jars/spark-core_2.12-3.3.0.jar) to field java.math.Big
WARNING: Please consider reporting this to the maintainers of org.apache.spark.util.SizeEstimator$
WARNING: Use --illegal-access=warn to enable warnings of further illegal reflective access operations
WARNING: All illegal access operations will be denied in a future release
Model loaded successfully!
Starting real-time detection (every packet triggers prediction)...
Try: curl google.com   OR   hping3 --flood ...   inside/outside the container
```

PRÉDICTION TEMPS RÉEL : BENIGN

```
youssef@wsoum:~/BigData_Project$ docker exec -it bigdata_project_spark-master_1 curl -s https://google.com > /dev/null
youssef@wsoum:~/BigData_Project$
```

```
================================================================================
 REAL-TIME DETECTION | 142.250.201.46:443 → 172.18.0.3:37928 (Proto: 6)
 PREDICTION → BENIGN | Attack Probability: 48.05%
================================================================================

================================================================================
 REAL-TIME DETECTION | 142.250.201.46:443 → 172.18.0.3:37928 (Proto: 6)
 PREDICTION → BENIGN | Attack Probability: 40.05%
================================================================================

================================================================================
 REAL-TIME DETECTION | 172.18.0.3:37928 → 142.250.201.46:443 (Proto: 6)
 PREDICTION → BENIGN | Attack Probability: 42.07%
================================================================================

================================================================================
 REAL-TIME DETECTION | 142.250.201.46:443 → 172.18.0.3:37928 (Proto: 6)
 PREDICTION → BENIGN | Attack Probability: 40.05%
================================================================================

================================================================================
 REAL-TIME DETECTION | 172.18.0.3:37928 → 142.250.201.46:443 (Proto: 6)
 PREDICTION → BENIGN | Attack Probability: 42.07%
================================================================================
```

ATTAQUE DÉTECTÉE : HPING3 FLOOD

```
youssef@wsoum:~/BigData_Project$ docker exec -it bigdata_project_spark-master_1 hping3 -S -p 80 --flood 8.8.8.8
HPING 8.8.8.8 (eth0 8.8.8.8): S set, 40 headers + 0 data bytes
hping in flood mode, no replies will be shown
```

```
================================================================================
REAL-TIME DETECTION | 172.18.0.3:1117 → 8.8.8.8:80 (Proto: 6)
PREDICTION → ATTACK | Attack Probability: 50.06%
MALICIOUS TRAFFIC DETECTED! BLOCK THIS FLOW NOW!
================================================================================

================================================================================
REAL-TIME DETECTION | 172.18.0.3:1118 → 8.8.8.8:80 (Proto: 6)
PREDICTION → ATTACK | Attack Probability: 50.06%
MALICIOUS TRAFFIC DETECTED! BLOCK THIS FLOW NOW!
================================================================================

================================================================================
REAL-TIME DETECTION | 172.18.0.3:1119 → 8.8.8.8:80 (Proto: 6)
PREDICTION → ATTACK | Attack Probability: 50.06%
MALICIOUS TRAFFIC DETECTED! BLOCK THIS FLOW NOW!
================================================================================

================================================================================
REAL-TIME DETECTION | 172.18.0.3:1120 → 8.8.8.8:80 (Proto: 6)
PREDICTION → ATTACK | Attack Probability: 50.06%
MALICIOUS TRAFFIC DETECTED! BLOCK THIS FLOW NOW!
================================================================================

================================================================================
REAL-TIME DETECTION | 172.18.0.3:1121 → 8.8.8.8:80 (Proto: 6)
PREDICTION → ATTACK | Attack Probability: 50.06%
MALICIOUS TRAFFIC DETECTED! BLOCK THIS FLOW NOW!
================================================================================
```

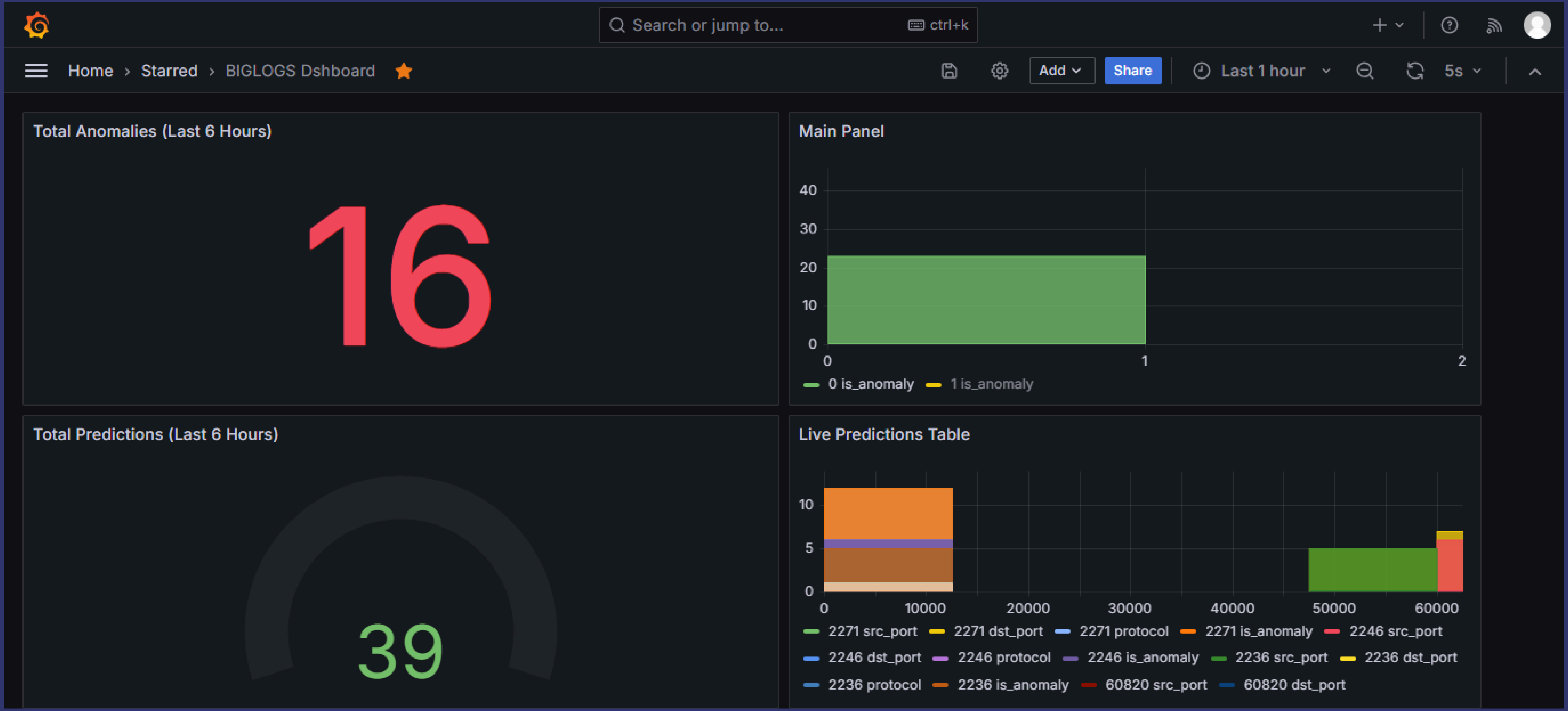
PRÉDICTIONS STOCKÉES CASSANDRA

```
youssef@wsoum:~/BigData_Project$ docker exec -it cassandra cqlsh cassandra 9042 -e "SELECT * FROM netflow.predictions;"
```

src_ip	ingestion_time	dst_ip	anomaly_score	dst_port	is_anomaly	protocol	src_port
87.120.191.124	2025-11-23 17:13:44.576000+0000	172.18.0.2	51.01613	8080	1	6	59414
87.120.191.124	2025-11-23 17:13:39.418000+0000	172.18.0.2	52.18523	8080	1	6	59414
172.18.0.6	2025-11-23 17:13:44.576000+0000	172.18.0.2	48.2189	7077	0	6	42434
172.18.0.4	2025-11-23 14:43:39.750000+0000	142.251.37.174	42.06936	443	0	6	37294
172.18.0.4	2025-11-23 14:43:36.927000+0000	142.251.37.174	43.06731	443	0	6	37294
172.18.0.4	2025-11-23 14:43:36.926000+0000	142.251.37.174	44.06731	443	0	6	37294
172.18.0.4	2025-11-23 14:43:34.166000+0000	142.251.37.174	42.06936	443	0	6	37294
172.18.0.4	2025-11-23 14:43:30.757000+0000	142.251.37.174	43.06731	443	0	6	37294
172.18.0.4	2025-11-23 14:43:27.220000+0000	142.251.37.174	44.06731	443	0	6	37294
172.18.0.4	2025-11-23 14:43:27.219000+0000	142.251.37.174	42.06936	443	0	6	37294
172.18.0.4	2025-11-23 14:43:23.081000+0000	142.251.37.174	42.06936	443	0	6	37294
172.18.0.4	2025-11-23 14:43:18.780000+0000	142.251.37.174	51.03295	443	1	6	37294
142.251.37.174	2025-11-23 14:43:39.750000+0000	172.18.0.4	48.05383	37294	0	6	443
142.251.37.174	2025-11-23 14:43:36.927000+0000	172.18.0.4	48.05383	37294	0	6	443
142.251.37.174	2025-11-23 14:43:36.926000+0000	172.18.0.4	50.05383	37294	1	6	443
142.251.37.174	2025-11-23 14:43:34.166000+0000	172.18.0.4	49.02134	37294	0	6	443
142.251.37.174	2025-11-23 14:43:30.757000+0000	172.18.0.4	49.05383	37294	0	6	443
142.251.37.174	2025-11-23 14:43:27.220000+0000	172.18.0.4	51.0317	37294	1	6	443
142.251.37.174	2025-11-23 14:43:23.081000+0000	172.18.0.4	40.05016	37294	0	6	443
142.251.37.174	2025-11-23 14:43:18.780000+0000	172.18.0.4	50.20768	37294	1	6	443
172.18.0.2	2025-11-23 17:13:44.576000+0000	87.120.191.124	51.05383	59414	1	6	8080
172.18.0.2	2025-11-23 17:13:39.418000+0000	87.120.191.124	53.20768	59414	1	6	8080

(22 rows)

Visualisation en Temps Réel Grafana



Conclusion

En conclusion, ce projet nous a permis de déployer un pipeline de cybersécurité Big Data entièrement fonctionnel, couvrant l'intégralité de la chaîne de traitement : de la capture réseau brute à l'analyse distribuée sous Spark, jusqu'à la visualisation intuitive pour les analystes via Grafana.

Nous avons atteint nos objectifs de performance sur les attaques testées, le tout soutenu par une architecture Docker résiliente et scalable.

Merci Pour Votre Attention